SOFTWARE ENGINEERING LAB REPORT

State Transition Diagram

LAB 7 | Case Study: Food Delivery Application

Project Overview

The Food Delivery Application allows customers to discover restaurants, browse menus, place and pay for orders, track delivery, and provide feedback. Restaurants manage menus and incoming orders, while delivery partners accept assignments and update delivery status. An administrator supervises users, restaurants, payments, and operational issues.

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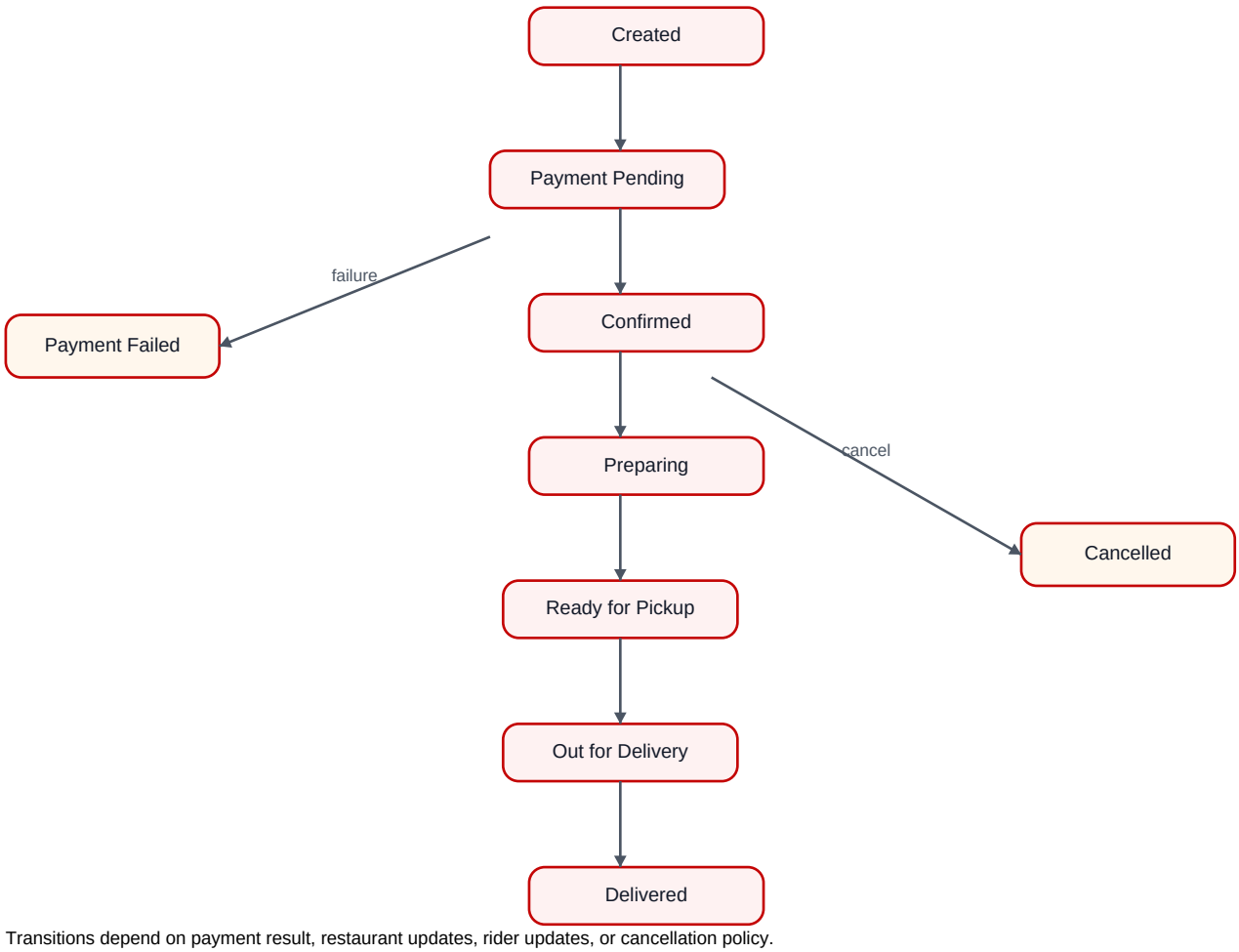
1. Aim

To describe the lifecycle of an order using a UML State Transition Diagram.

2. Order States

State	Meaning
Created	Order request is recorded but not yet paid/confirmed.
Payment Pending	Payment authorization or confirmation is in progress.
Confirmed	Payment is successful and order is sent to restaurant.
Preparing	Restaurant is preparing the selected items.
Ready for Pickup	Order is packed and ready for rider collection.
Out for Delivery	Delivery partner has collected the order.
Delivered	Customer has received the order; terminal success state.
Payment Failed	Payment did not complete; retry or abandon.
Cancelled	Order was cancelled under the applicable policy.

3. State Transition Diagram



Transitions depend on payment result, restaurant updates, rider updates, or cancellation policy.

4. Transition Rules

• Created → Payment Pending: customer submits checkout.

• Payment Pending → Confirmed: payment provider reports success.

• Payment Pending → Payment Failed: payment is declined, times out, or fails.

• Confirmed → Preparing → Ready for Pickup → Out for Delivery → Delivered: restaurant and rider progress updates.

• Eligible non-terminal states may transition to Cancelled according to policy; if money was captured, the system should initiate and track a refund.

5. Result

The state diagram defines the order lifecycle and clarifies which status updates are expected from payment, restaurant, and delivery subsystems.